

WHO NEEDS WHAT?

- **Office/Student Users:** Ensure you have at least 2 x USB-A (for pen drives, mouse etc.), an HDMI (for connecting to projectors/TVs in meeting rooms or class), and a headphone jack. Most midrange laptops like HP Pavilion or Lenovo IdeaPad provide these. USB-C with charging is nice to have (for universal charger use), and if it supports DisplayPort you can use it for a second monitor as well.
- **Creative Professionals:** You'll want Thunderbolt/USB4 for high-speed external drives (video editors, scratch disks) and for hooking up multiple monitors. An SD card reader is very important for photographers. HDMI 2.1 or DP can be useful if you're driving a 4K reference monitor at 60+ Hz. Consider laptops like MacBook Pro 16 (M3) or Gigabyte Aero which come with a rich set of ports (Thunderbolt, HDMI 2.1, SD UHS-II) – so you can plug your camera card and two external displays without adapters. If your chosen ultrabook lacks some ports, budget for adapters (e.g., a USB-C hub that has SD, HDMI, and USB-A).
- **Gamers:** HDMI 2.1 is highly desirable if you intend to game on a big 4K TV or a new monitor at high refresh. Multiple USB-A ports are useful for your gaming mouse, keyboard, gamepad, etc. (gaming laptops like ASUS TUF or MSI have 3 or more USB-A typically). Also, an RJ-45 Ethernet port is a plus for reducing lag. If the gaming laptop supports USB-C charging (some do via USB-PD when not gaming), that's a bonus for portability but you'll still use

the heavy AC adapter when playing. Thunderbolt isn't critical for pure gaming, but it's nice for future expansion (e.g. external GPU or fast storage).

- **Traveling Professionals:** If you travel, you might prefer fewer ports on the device and rely on a travel dock for when you need them – this keeps the laptop light. But ensure at least two USB-C ports so you can charge and connect another device simultaneously. If giving lots of presentations, HDMI is your friend (most conference rooms have HDMI hookups). A business-travel-focused model like Lenovo ThinkPad X1 Carbon typically has 2 x Thunderbolt, 1 x USB-A, HDMI, and a headphone jack – a balanced combo for work on the go.
- **Who can live with just USB-C?** Some new laptops (e.g. certain thin Asus ZenBooks, HP Dragonfly models) come only with USB-C ports (and maybe a headphone jack). If you're a tech minimalist and everything you use is wireless or USB-C, this can work. For example, newer smartphones and SSDs often use USB-C cables; monitors can connect via USB-C; Bluetooth handles mice and audio. You'd likely get a small USB-C multiport adapter for the occasional legacy connection. It's doable, but think about your devices: Do you often plug in USB drives (you'd need an adapter)? Do you connect to old projectors (you'd need a USB-C to VGA/HDMI dongle)? If those scenarios are frequent, a laptop with built-in HDMI and USB-A will save you a lot of fuss.

- **HDMI:** This is the most common video-out port, handy for connecting to external monitors, TVs, or projectors. The key is the HDMI version:
 - **HDMI 2.0** supports up to 4K at 60Hz. This is fine for a typical 4K TV or a standard office monitor. Most laptops from the past few years have at least HDMI 2.0.
 - **HDMI 2.1** supports 4K at 120Hz, 8K, and other features. If you plan to connect to a high-end 4K 120Hz monitor or TV (say for gaming on an OLED TV at 120Hz), you need HDMI 2.1 or alternatively use DisplayPort over USB-C. Many gaming laptops started including HDMI 2.1 around 2021–2022 to allow this high-bandwidth output. For example, Razer Blade and Gigabyte Aorus models advertise HDMI 2.1 so you can game on a big 4K 120Hz TV properly. Some creator laptops (like the Gigabyte Aero) also offered HDMI 2.1 to connect to reference monitors or multiple displays. If it's HDMI 2.0, you're limited to 4K60 (or 1440p at 144Hz, etc.).

- For most users, HDMI 2.0 is adequate: you can do presentations on projectors, watch UHD content on a TV, etc. But if you know you want to push higher refresh rates or future-proof for an 8K display, look for HDMI 2.1. Keep in mind that even if a laptop's GPU could technically output higher, the physical port needs to support it. Some laptops with powerful GPUs only had HDMI 2.0 due to internal routing, so it's worth double-checking. Business laptops sometimes stick to 2.0 to save power, whereas gaming ones go 2.1.
- **DisplayPort:** You won't often find a full-size DisplayPort jack on a laptop (some large gaming models or mobile workstations include one). Instead, DisplayPort is typically available through USB-C (via Alt Mode). If you have a high-res, high-refresh monitor with only DisplayPort (common for QHD 165Hz panels, etc.), you can use a USB-C to DP cable/dongle, provided the laptop's USB-C supports DP Alt Mode. Thunderbolt docks also often give you DisplayPort outputs. So lack of a physical DP port is usually not a problem if you